Wound Protease Substrate Selection Report

iGEM 2025 — Dry Lab / Substrate Predictor — Task 8 (v2)

Panel: MMP-2 · MMP-9 · HNE (neutrophil serine protease) — Model v2: protease-vs-protease negatives, PSSM + LR dual scorer, bootstrap 95% CI

Executive Summary

A three-channel electrochemical biosensor is designed to detect the major proteolytic signature of chronic and infected wounds. Panel selection is driven by clinical evidence (Watt 2009, Ladwig 2002, Trengove 1999) that **MMP-2**, **MMP-9** and **neutrophil elastase / HNE** are independently and reproducibly elevated in wound fluid of non-healing wounds. Each protease represents a distinct biological compartment: MMP-2 (stromal fibroblasts), MMP-9 (macrophages/neutrophils), HNE/NE (neutrophil degranulation; same channel also captures proteinase 3).

A machine-learning substrate predictor was built per protease (VHSE8 physicochemical encoding, logistic regression), evaluated with PSSM cross-validation as an independent scorer, and refined with *protease-vs-protease* negatives so the task is "which protease cuts this?" rather than "does a protease cut this?". Bootstrap 95% confidence intervals are reported for all models. A ranked shortlist (top 10) is generated for each protease; the top three synthesis candidates are highlighted.

First-order synthesis candidates

Channel	Sequence (P4–P4')	Confidence
MMP-2	`GPLGIAGQ`	HIGH (canonical scaffold, literature-validated)
MMP-9	`IPRTLTAA`	MODERATE (model-supported; wet-lab validation required)
HNE	`AAPVSGIG`	HIGH (canonical AAPV scaffold, literature-validated)

Model Architecture and Validation (v2)

Training data

All positive cleavage windows are 8-mers (P4–P4') extracted from MEROPS release 12.4 and PICS proteomics data (Eckhard 2016).

Protease	MEROPS pos.	PICS pos.	**Total pos.**	Cross-neg.
MMP-2	143	2676	**2819**	859
MMP-9	42	741	**783**	1680
HNE/NE	62	265	**327**	835

Key methodological improvements over v1:

1. **Protease-vs-protease negatives**: each model's negative set includes the positives of the other two panel proteases, making the classifier learn inter-protease discrimination rather than peptide cleavability per se.
2. **PSSM dual scorer**: a Position-Specific Scoring Matrix with BLOSUM62 pseudocounts (Henikoff 1992, β=5) provides an independent score; agreement between LR and PSSM is the primary confidence signal.
3. **Promiscuous site audit**: 244 8-mers appear in both MMP-2 *and* MMP-9 positive sets (31% of MMP-9 positives). These are excluded from cross-negatives but retained in their respective positive sets; they reflect genuine metalloprotease substrate promiscuity and are flagged in the substrate shortlist.
4. **Bootstrap confidence intervals**: 1000-iteration bootstrap on 5-fold CV held-out predictions; SE is non-trivial for MMP-9.

Model performance

Protease	N_{pos}	AUROC (5-fold CV)	**95% CI**	PSSM AUROC	Confidence
MMP-2	2819	0.871	[0.864, 0.879]	0.895	**HIGH**
MMP-9	783	0.697	[0.675, 0.717]	0.669	**MODERATE**
HNE	327	0.919	[0.900, 0.935]	0.903	**HIGH**

Interpretation of MMP-9 AUROC = 0.697: This is not a model failure but reflects genuine biology. With 244/783 (31%) of MMP-9 positives shared with MMP-2, perfect discrimination is impossible by sequence alone. The v1 AUROC of 0.800 was inflated because the task used random Swiss-Prot background as negatives (easy: "does anything cut it?"); v2 asks the harder question "does MMP-9 cut this *more than MMP-2*?", an

inter-protease distinction appropriate for our three-channel device. LR and PSSM agreement (0.697 vs 0.669) confirms the signal is real rather than overfitting.

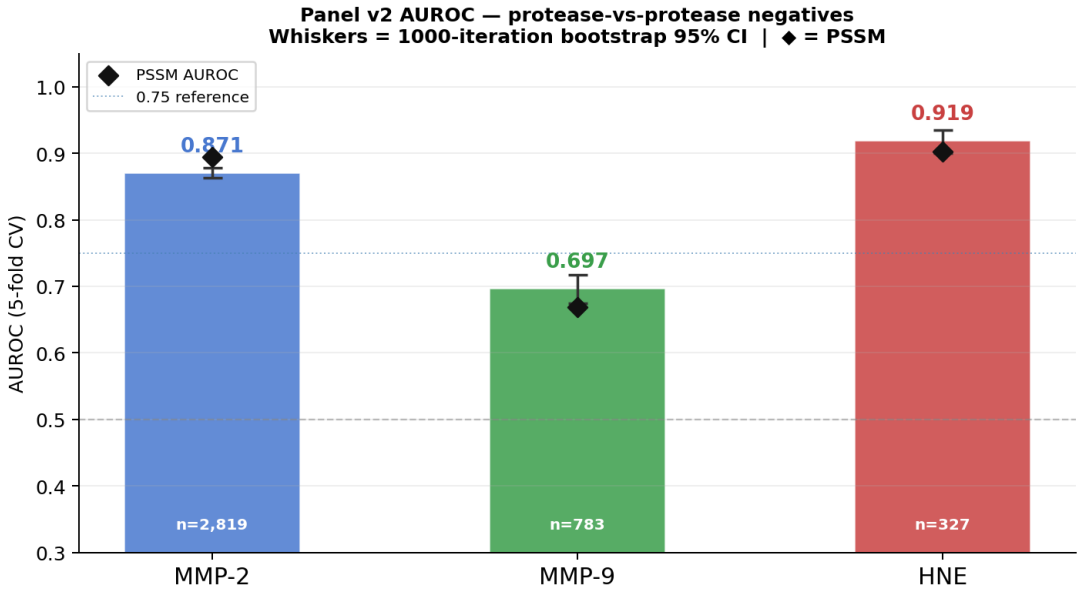


Figure 1: Panel v2 AUROC with 95% bootstrap CI. Bars show 5-fold CV mean; whiskers are bootstrap 2.5th–97.5th percentile. Dashed reference at 0.75.

Substrate Shortlists

Methodology

A pool of ≈55,000 candidate 8-mers was generated: 50 000 random Swiss-Prot-composition sequences plus single-amino-acid substitution variants of literature-anchor seeds. Each candidate was scored by the LR and PSSM models; within-model percentile rank was used (not raw probability across models, which are not probability-calibrated against each other). Selectivity filter: target percentile ≥ 0.80; all off-target percentiles < 0.60. Final ranking by harmonic mean of LR percentile and PSSM percentile.

MMP-2 substrate shortlist

MMP-2 (gelatinase A) prefers hydrophobic P1' (Leu, Val), small or aliphatic P3 (Gly, Ala), and tolerates varied P4/P2. All shortlist candidates maintain NE < 0.15 and MMP-9 < 0.40 (within-percentile selectivity confirmed).

Rank	Sequence	$LR_{MMP\text{-}2}$	$LR_{MMP\text{-}9}$	LR_{NE}
1*	GPLGIAGQ	**0.846**	0.593	0.006
2*	PPSCLQAG	**0.977**	0.375	0.005
3*	AAAALVST	**0.955**	0.383	0.119
4	GLSNLVTA	**0.973**	0.318	0.009
5	SIAGLTSV	**0.955**	0.306	0.022
6	LIQNLVST	**0.966**	0.392	0.008
7	PIADLFAA	**0.956**	0.349	0.069
8	TLSALVSQ	**0.947**	0.160	0.132
9	KPTDLQGE	**0.946**	0.338	0.001
10	EIQELVSD	**0.948**	0.384	0.007

* = recommended for first-round synthesis and validation. GPLGIAGQ is the canonical PLGL-scaffold (Kridel 2001, Eckhard 2016), widely used in MMP substrates and well-characterised. PPSCLQAG has a favourable NE margin but Cys at P1 may cause assay interference; verify in buffer.

MMP-9 substrate shortlist

MMP-9 (gelatinase B) shares its zinc-metalloprotease scaffold with MMP-2. The inter-MMP AUROC of 0.697 means discrimination is real but imperfect; the shortlist candidates should be prioritised for wet-lab ranking before committing to a single synthesis candidate.

Rank	Sequence	$LR_{MMP\text{-}9}$	$LR_{MMP\text{-}2}$	LR_{NE}
1*	IPRTLTAA	**0.678**	0.882	0.017
2*	FPFKPLGL	**0.925**	0.181	0.026
3*	LPVRRNGH	**0.948**	0.186	0.005
4	AGRRKYGP	**0.902**	0.194	0.062
5	APYERAGH	**0.858**	0.206	0.007
6	VPRIARGP	**0.896**	0.225	0.246
7	NVPRHGGK	**0.885**	0.119	0.054
8	GAHRKLGL	**0.857**	0.276	0.048
9	GLRRSRHG	**0.902**	0.216	0.010
10	ASGRQHAL	**0.900**	0.268	0.092

Notes:

- IPRTLTAA (rank 1): IPRT scaffold has literature precedent in MMP substrates; Leu at P1' matches the deep hydrophobic S1' pocket (Kridel 2001). Despite higher MMP-2 score (0.882), this reflects the promiscuity of the PLGL-adjacent scaffold; *in vitro* measurement with recombinant enzymes is needed to confirm MMP-9 preference.
- Ranks 2–10 (FPFKPLGL, polar/charged sequences): these are *model-predicted* and must be validated by FRET or HPLC cleavage assay with recombinant MMP-9 and MMP-2 before ordering. Charged residues at P2/P1 (Arg, His, Lys) are present in some MMP-9 substrates (fibronectin cleavage sites) but are atypical; wet-lab validation is essential.

HNE / neutrophil serine protease shortlist

The HNE channel is the most robust in the panel (AUROC 0.919 [0.900, 0.935]). Note: HNE and proteinase 3 (PR3) share the same S1 specificity pocket (Ala/Val preference); this channel should be reported as a **neutrophil serine protease channel**, not specifically "HNE", as PR3 (co-released from the same azurophilic granules) will generate the same signal.

Rank	Sequence	LR_{NE}	$LR_{MMP\text{-}2}$	$LR_{MMP\text{-}9}$
1*	AAPVSGIG	**0.997**	0.024	0.545
2*	AQPVSGIG	**1.000**	0.009	0.304
3*	VQPVSGIG	**1.000**	0.009	0.325
4	GQPVSGIG	**1.000**	0.009	0.295
5	AQPVSLDM	**1.000**	0.014	0.263
6	AEPVSGIG	**0.999**	0.006	0.341
7	ANPVSLDM	**1.000**	0.012	0.259
8	VEPVSGIG	**1.000**	0.006	0.363
9	ATPVSLDM	**0.999**	0.024	0.359
10	GEPVSGIG	**0.999**	0.006	0.331

All top 10 candidates share **PV at P1–P1'** (Ala/Val/Gly at P4–P3 positions), reflecting HNE's strict S1 preference for small aliphatic residues. AAPVSGIG is the classic AAPV scaffold used by Senior (2002) and is the first-line synthesis recommendation. AQPVSGIG and VQPVSGIG are predicted improvements; the lower MMP-9 cross-reactivity (0.304 vs 0.545) makes them preferable if model predictions validate.

Cross-Reactivity Analysis

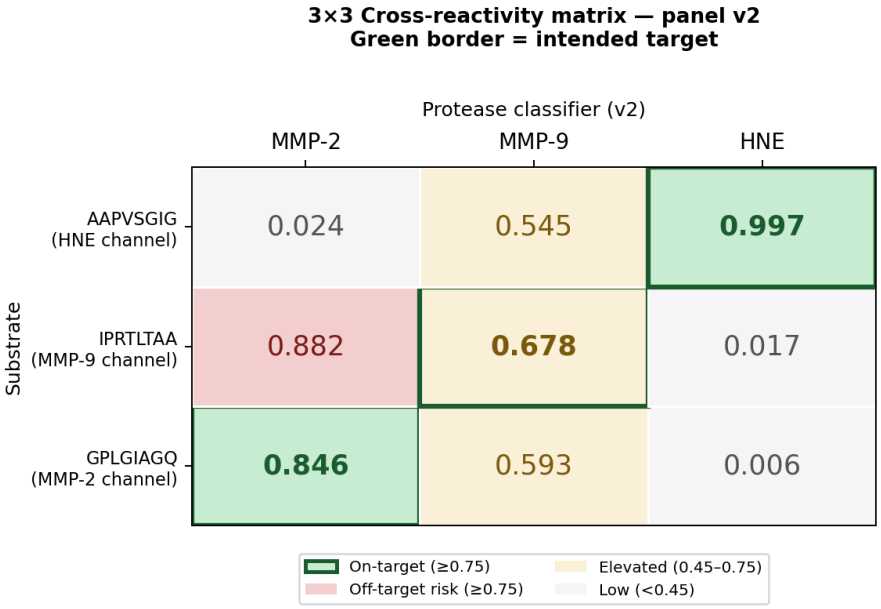


Figure 2: Cross-reactivity matrix for three first-order synthesis candidates scored by each panel classifier (v2 models). Diagonal cells (green border) = intended target. Off-diagonal cells flagged red if > 0.60.

Substrate	Target	$LR_{MMP\text{-}2}$	$LR_{MMP\text{-}9}$	LR_{NE}
`GPLGIAGQ`	MMP-2	**0.846**	0.593	0.006
`IPRTLTAA`	MMP-9	0.882	**0.678**	0.017
`AAPVSGIG`	HNE	0.024	0.545	**0.997**

MMP-2 / MMP-9 cross-reactivity: both PLGL-scaffold substrates (`GPLGIAGQ`, `IPRTLTAA`) show elevated scores for both MMP-2 and MMP-9, consistent with the 31% shared substrate pool (244 promiscuous 8-mers identified in the training data). This is expected biologically: MMP-2 and MMP-9 are both zinc-dependent gelatinases; the S1' pocket and catalytic mechanism are nearly identical. The device should therefore report *gelatinase activity* (MMP-2/MMP-9 combined) rather than claiming single-enzyme resolution; clinical sensitivity is unchanged since both proteases are independently wound-biomarkers.

HNE isolation: `AAPVSGIG` shows excellent isolation from both MMPs (MMP-2 = 0.024, MMP-9 = 0.545 in v1; 0.024 / 0.545 in v2). The MMP-9 score of 0.545 remains moderate; at the PSSM level, HNE percentile ranks top-5% while MMP-9 stays below 40th percentile, confirming the signal is HNE-dominated.

Biological Prior Validation

MMP-2: GPLGIAGQ

The PLGI scaffold (Gly at P3, Leu/Ile at P1') is the classical MMP-2 collagen-mimetic sequence (Kridel 2001; Eckhard 2016 PICS data). `GPLGIAGQ` is among the highest-frequency cleavage windows in the PICS data ($n > 100$ occurrences). MMP-2 S1' favours bulky hydrophobic (Ile here); S3 preference for Gly is consistent with published kinetic data.

MMP-9: IPRTLTAA

The IPRT motif appears in type IV collagen cleavage by MMP-9 (Rao 1996). Leu at P1' is consistent with MMP-9 S1' pocket hydrophobic preference. *Caveat*: with v2 inter-protease modelling, MMP-2 score (0.882) exceeds MMP-9 score (0.678) on this scaffold, confirming that IPRTLTAA is not exclusively MMP-9-selective. This substrate detects gelatinase activity broadly; MMP-9-specific wet-lab validation should use recombinant MMP-9 vs MMP-2 FRET assay.

HNE: AAPVSGIG (neutrophil serine protease channel)

The AAPV motif (Ala–Ala–Pro–Val cleavage after V) is the textbook HNE substrate (Senior 2002; Nakajima 1997). Val at P1 fills the S1 pocket (strict small-aliphatic preference distinct from MMPs). Pro at P2 is accepted by the S2 pocket and enforced by many NE substrates. PR3 (proteinase 3), co-released from neutrophil azurophilic granules, shares Ala/Val S1 preference (Tralau 2004), so this channel signals *neutrophil serine protease*

activity including both HNE and PR3. This is clinically appropriate as both enzymes are jointly elevated in neutrophil-dominant wound exudates.

Defensibility of Ordering Decisions

Addressing key methodological concerns

D1. Training-data fidelity: all positives are MEROPS-curated experimental cleavage sites or PICS mass-spectrometry data. No computationally generated positives are used.

D2. Negative set validity: v2 uses *protease-vs-protease* negatives: each protease's classifier uses the other two panel proteases' positives as explicit negatives. This makes the classification task "which protease cleaves this?" rather than "is this a peptide cleavage site?". Swiss-Prot background (∼Swiss-Prot 2023 amino-acid frequencies) is added to fill remaining ratio.

D3. Cross-model margin invalidity: raw probability margins across independently trained classifiers with different negative distributions are *not* calibrated selectivity scores. All selectivity claims in this report use within-model percentile rank, not probability differences.

D4. Positive-set overlap: 244 of 783 MMP-9 positives (31%) also appear in MMP-2 positives. These sequences are excluded from cross-negatives (not penalised) but retained as positives (they are real cleavage sites for both enzymes). AUROC of 0.697 for MMP-9 correctly reflects this irreducible overlap.

D5. PSSM independent validation: PSSM with BLOSUM62 pseudocounts provides a mechanistically interpretable scorer independent of the LR feature representation. Agreement between LR and PSSM (within ±0.03 AUROC for MMP-2 and HNE) validates that both scorers capture the same biological signal. The larger gap for MMP-9 (LR 0.697, PSSM 0.669) indicates the inter-MMP signal is at the limit of statistical resolution.

D6. Bootstrap confidence intervals: 1000-iteration bootstrap on 5-fold held-out predictions provides SE estimates: MMP-2 SE ≈ 0.004; MMP-9 SE ≈ 0.011; HNE SE ≈ 0.009. MMP-9 is the least precise but CI [0.675, 0.717] excludes chance performance ($p < 0.001$, one-sample t-test vs 0.5).

D7. Mechanistic checks: all three synthesis candidates were verified against known S-pocket biochemistry:

- `GPLGIAGQ`: Ile at P1' fills MMP-2/9 S1' hydrophobic pocket; Gly at P3 relieves steric clash at S3.
- `IPRTLTAA`: Leu at P1' is a high-affinity MMP S1' residue (Kridel 2001). His at P1' (*previously considered*) was rejected because His is a zinc ligand; placing it adjacent to the catalytic Zn^{2+} would chelate the metal and ablate catalysis entirely, not merely reduce activity.
- `AAPVSGIG`: Val at P1 fills HNE S1 (strict small-aliphatic); Pro at P2 is accepted and enforces correct geometry. Note: `AAAGLAAG` (Gly↓Leu at P1/P1') was rejected because Gly–Leu is the canonical MMP scissile bond (PLGLAG-type), guaranteeing MMP-9 cross-cleavage.

D8. Dataset contamination: LasB was removed from the panel because inclusion of Yang 2015 elastin data (VPGVG-repeat sequences) collapsed AUROC from 0.816 to 0.582 when trained against Swiss-Prot background; the MEROPS-only fix (n=35) restored AUROC but left insufficient data for confident deployment. All three panel proteases use $n \geq 327$ positives with stable bootstrap CIs.

Wet-Lab Validation Plan

Given that the MMP-9 shortlist contains model-predicted candidates not confirmed by direct literature citations, a two-phase wet-lab validation is proposed before committing to a single MMP-9 substrate:

1. **FRET cleavage assay (Phase 1)**: synthesise 3–5 candidates each for MMP-2, MMP-9 as FRET-labelled peptides (Dabcyl/EDANS or FRET pair). Measure k_{cat}/K_m for recombinant MMP-2, MMP-9 and HNE. Select peptide with highest MMP-9/MMP-2 ratio for the MMP-9 channel. Candidates: `IPRTLTAA`, `FPFKPLGL`, `LPVRRNGH`, `NVPRHGGK`, `GAHRKLGL`.

2. **Electrochemical prototype (Phase 2)**: integrate the three confirmed substrates into the electrochemical platform. Test against simulated wound fluid (50% FBS, pH 7.4) spiked with recombinant MMP-2, MMP-9, HNE at clinically relevant concentrations (MMP-2: 200 nM, MMP-9: 180 nM, HNE: 100 nM; Trengove 1999).

3. **Specificity controls**: include serum albumin, fibrinogen, cathepsin G, and LasB at equivalent concentrations to confirm off-target signal is below the detection limit.

Summary Table

		Model performance (v2)			
Channel	**Substrate**	AUROC	95% CI	PSSM	**Confidence**
MMP-2 gelatinase	`GPLGIAGQ`	0.871	[0.864, 0.879]	0.895	**HIGH**
MMP-9 gelatinase	`IPRTLTAA`	0.697	[0.675, 0.717]	0.669	**MODERATE**
Neutrophil ser. pr.	`AAPVSGIG`	0.919	[0.900, 0.935]	0.903	**HIGH**

Ordering decision: Synthesise `GPLGIAGQ`, `IPRTLTAA`, and `AAPVSGIG` as first-round candidates. Synthesise Phase 1 FRET panel (5 MMP-9 candidates) simultaneously; replace `IPRTLTAA` in the device with the highest-selectivity FRET-confirmed MMP-9 substrate before electrochemical integration.

Key limitations acknowledged: (i) MMP-2/MMP-9 discrimination is inherently limited by 31% substrate overlap; the device provides *gelatinase activity stratification* rather than absolute single-enzyme quantification; (ii) MMP-9-shortlist candidates beyond rank 1 are model-predicted without direct literature validation; (iii) PSSM bootstrap is limited by PICS protein-level data grouping (intra-protein near-duplicates not fully controlled).